Day 2: TCP/IP Suite

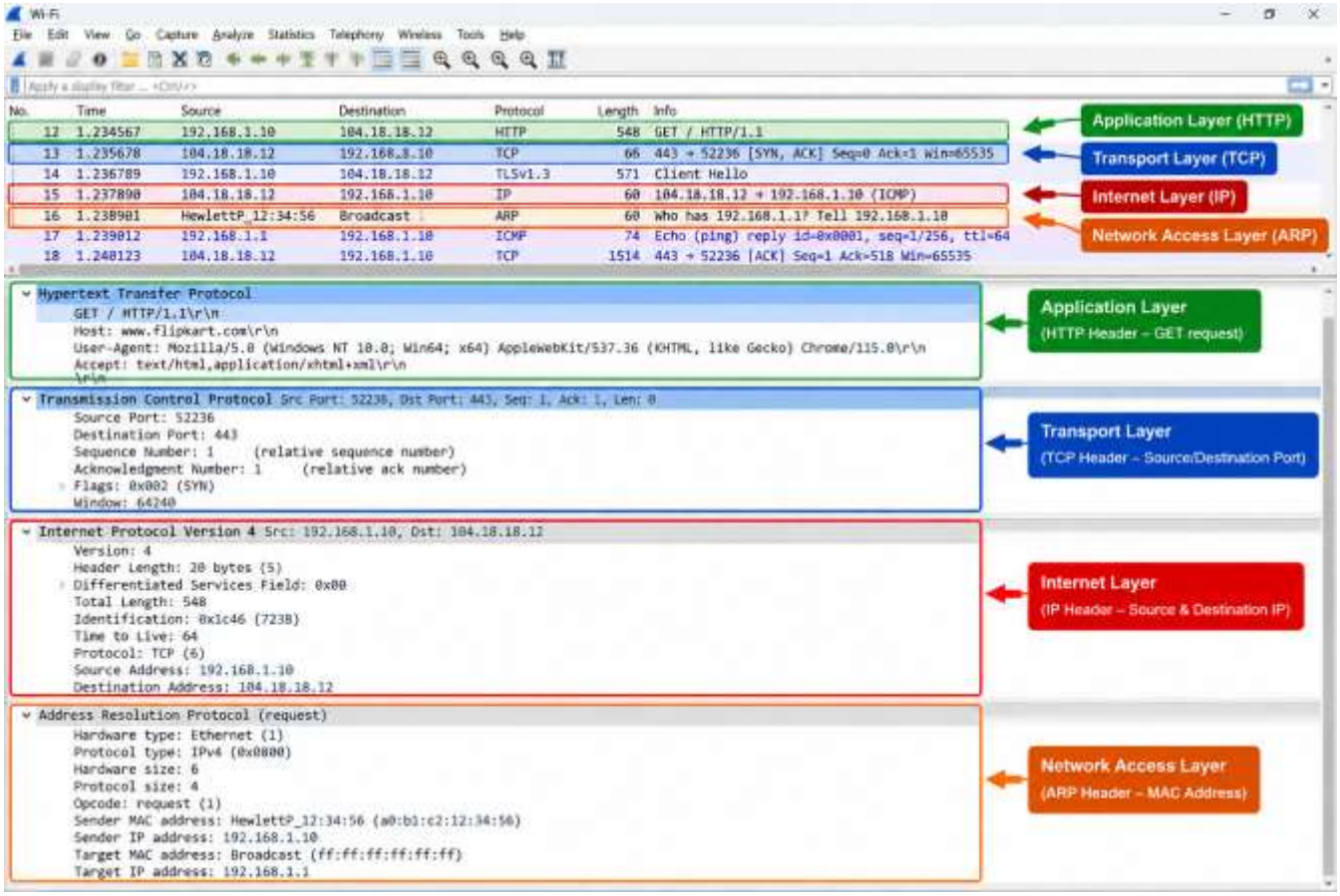
Practical Task Answers

Q1. Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.

Steps followed

1. Open Wireshark and select the main interface (Wi-Fi or Ethernet). Click the shark-fin button to start the capture.
2. Open flipkart.com in the browser and refresh the page (press F5). Wait until it loads, then stop the capture with the red square button.
3. Click on a packet in the list and expand its headers in the Packet Details pane. Each header belongs to one TCP/IP layer.

Screenshot of my capture:



One packet for each TCP/IP layer

TCP/IP layer	Packet No.	Typical packet	Header to expand and fields to note	What it shows
Application	____	DNS: Standard query A www.flipkart.com	Domain Name System (query): Queries, Name = www.flipkart.com, Type A	The app asks for the IP address of Flipkart's website.

TCP/IP layer	Packet No.	Typical packet	Header to expand and fields to note	What it shows
Transport	____	TCP: 5xxxx to 443 [SYN]	Transmission Control Protocol: Source Port, Destination Port 443, Sequence Number, Flags (SYN), Window Size	A reliable connection is being started with the secure web port.
Internet	____	Any IPv4 packet to Flipkart's server	Internet Protocol Version 4: Source Address (my laptop's IP), Destination Address (Flipkart server), TTL, Protocol (TCP = 6)	The IP header decides where the packet goes across networks.
Network Access	____	Any frame (or an ARP packet)	Ethernet II: Source MAC (my laptop), Destination MAC (my router), Type (IPv4 0x0800)	The frame is delivered to the next device on the local network.

Note: Every packet carries several headers, one inside the other. A packet is chosen for a layer according to its main protocol shown in the Protocol column. Fill in the Packet No. from your capture. Flipkart uses HTTPS, so the page content appears as encrypted TLS data.

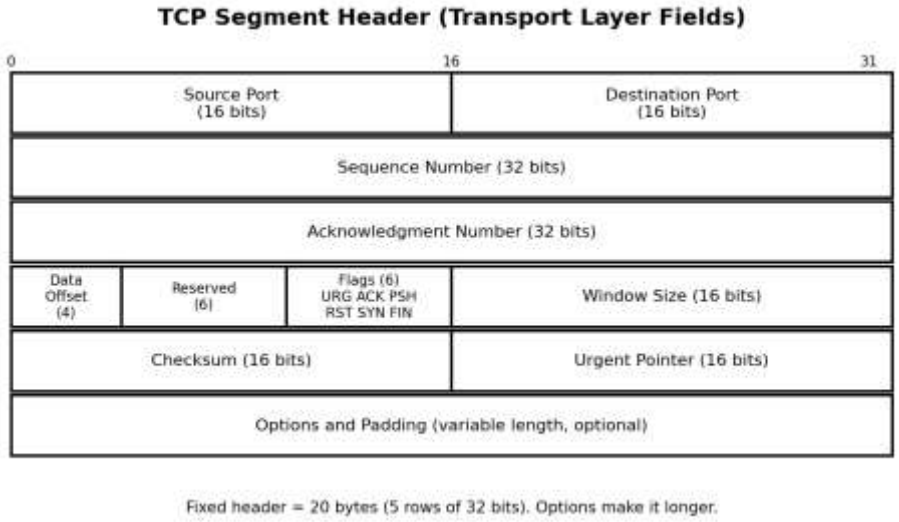
Q2Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).

OSI layer	TCP/IP layer	Function	Protocols / devices	Real-world example
7 Application	Application	Gives network services to apps	HTTP, HTTPS, DNS, SMTP, FTP	Sending a WhatsApp message
6 Presentation	Application	Encryption, compression, data formats	TLS, JPEG, ASCII	Instagram photo shown as JPEG
5 Session	Application	Starts, keeps and ends sessions	NetBIOS, RPC, sockets	Staying logged in to Zomato
4 Transport	Transport	Reliable (TCP) or fast (UDP) delivery with ports	TCP, UDP	Zomato order delivered fully with TCP
3 Network	Internet	IP addressing and routing	IP, ICMP, router	Router sending a Flipkart request to the right server
2 Data Link	Network Access	Frames, MAC addresses, error check	Ethernet, WiFi, switch	WiFi router sending data to my phone
1 Physical	Network Access	Sends bits as signals	Cables, hub, radio signals	WiFi signal or fibre line carrying data

Main differences

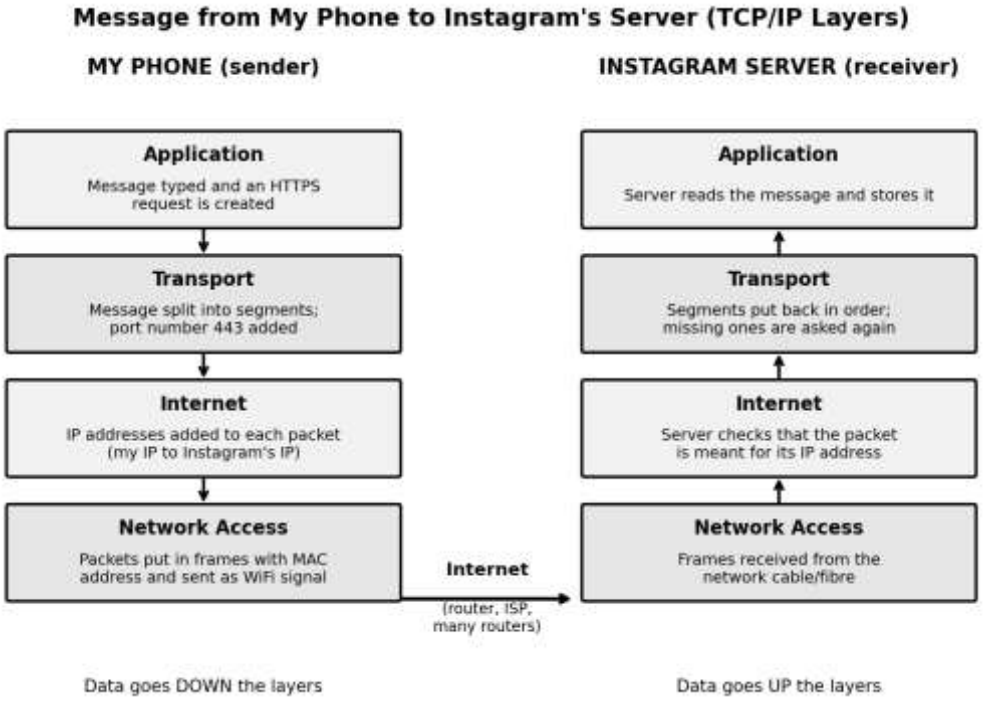
- The OSI model has 7 layers and the TCP/IP model has 4 layers (the top 3 OSI layers are joined into one Application layer, and the bottom 2 into one Network Access layer).
- OSI is a theoretical reference model used for learning and troubleshooting; TCP/IP is the practical model on which the internet actually works.
- In TCP/IP the Application layer handles the work of presentation and session, so there are no separate layers for them.

Q3. Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines. Look for fields like Source Port, Destination Port, Sequence Number, etc.



Field	Size	What it does
Source Port	16 bits	The port number of the sending application (for example 52341, chosen randomly by my browser). It tells the receiver where to send the reply.
Destination Port	16 bits	The port number of the service on the server (for example 443 for HTTPS or 80 for HTTP). It tells which application should get the data.
Sequence Number	32 bits	Number of the first byte of data in this segment. It lets the receiver put segments in the correct order.
Acknowledgment Number	32 bits	The number of the next byte the receiver expects. It confirms that the earlier data was received correctly.
Data Offset (Header Length)	4 bits	Shows how long the TCP header is, so the receiver knows where the data starts.
Reserved	6 bits	Kept for future use. It is set to zero.
Flags	6 bits	Control bits: SYN starts a connection, ACK acknowledges data, FIN ends a connection, RST resets it, PSH pushes data immediately and URG marks urgent data.
Window Size	16 bits	How much data the receiver can accept at one time. It is used for flow control so the sender does not overload the receiver.
Checksum	16 bits	Used to detect errors in the header and data. If it does not match, the segment is discarded and sent again.
Urgent Pointer	16 bits	Points to urgent data inside the segment. It is used only when the URG flag is set.
Options and Padding	Variable	Extra settings such as the maximum segment size. Padding makes the header length a multiple of 32 bits.

Q4. Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).



TCP/IP layer	What happens (in simple terms)
Application	I type a message or tap Post in Instagram. The app creates an HTTPS request for Instagram's server.
Transport	The message is split into small segments and numbered. TCP adds the port numbers (my random port and 443) and makes sure every piece arrives.
Internet	An IP header is added to each segment. It contains my phone's IP address and Instagram's server IP address. Routers use it to choose the path.
Network Access	Each packet is put in a frame with the MAC address of my WiFi router. It is sent as WiFi radio signals and then through cables and fibre of the ISP.
At Instagram's server	The same steps happen in reverse order: frames are received, the IP header is checked, segments are joined in order, and the Instagram app on the server reads and saves the message.

Q5. Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.

Packet chosen: a TCP packet (Packet No. ____, replace with the number from your capture) with the Info column "52341 to 443 [SYN]", seen when I opened the BookMyShow page.

Point	Answer
TCP/IP layer	Transport layer.
How I identified it	The Protocol column shows TCP. In the details pane the header Transmission Control Protocol appears, with the source port, the destination port 443 and the SYN flag. Ports, sequence numbers and flags are Transport layer fields.
Real-world action	It is the first step (SYN) of a connection started by my browser or app when I open BookMyShow or tap Book Tickets. The device is asking the server to open a secure connection before any movie data is sent.

After the SYN packet, the server replies with SYN-ACK and my device sends ACK (the three-way handshake). Only after this does the encrypted data travel, so the seat list and payment details are sent safely.